

Train-Free Visual Numerosity Diagnostics Under a Strict Evidence Gate

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Abstract

Exact visual counting is a useful stress test for multimodal models because a single integer answer depends on category grounding, instance separation, visual aggregation, and output parsing, yet many evaluations blur exact-count failures with dataset scope, parser behavior, or unavailable baselines. We report a train-free visual numerosity diagnostic study over CAPTURE real validation, TallyQA, and PixMo-Count using one fixed hosted multimodal snapshot as the single evaluated backend, a single-integer response contract, deterministic integer extraction, exact match and mean absolute error scoring, eight prompt or decomposition conditions, three repeats, 140 ready tasks, and 7140 image requests. The measured result is negative rather than a method win: 3360 scored rows yield overall exact match 0.158631, and CAPTURE real collapses to 0.040833 exact match with 22.422083 mean absolute error. Parser and empty-response failures are absent; 2827 rows are parsed numeric answers that are simply wrong, while secondary VLM, FSC-147, and object-aware detector/decomposition conditions remain fail-closed because the required backend, image assets, or detector resources are not verified. The evidence supports a diagnostic result about weak cross-family exact counting and numeric error modes, not a broad positive crop/grid improvement claim.

Introduction

Visual counting looks narrow because the output is a single integer, but an exact answer requires several operations to line up. A model must identify the requested class, decide which visible instances count, separate adjacent or occluded objects, aggregate across the image, and return a parseable number. Psychophysical work already separates small-number subitizing from larger exact enumeration (Kaufman et al. 1949; Trick and Pylyshyn 1994; Jevons 1871; Piazza et al. 2004). Modern vision-language models add instruction following, multimodal grounding, and free-form generation to that chain (Radford et al. 2021; Alayrac et al. 2022; Li et al. 2023; Liu et al. 2023a; Wang et al. 2025).

The motivating question was whether train-free decomposition can improve exact counting relative to matched full-image prompting. Prior counting and grounding work makes that question plausible: crowd-counting datasets, open-world counters, and divide-and-conquer visual-language counting methods all suggest that local evidence can matter (Zhang et al. 2016; Ranjan et al. 2021; Liang et al.

2023; Amini-Naieni, Han, and Zisserman 2024; Qharabagh, Ghofrani, and Fountoulakis 2024; Chen et al. 2026). At the same time, VQA-style and compositional benchmarks show that the same integer answer format can hide different grounding and selection requirements (Agrawal et al. 2016; Krishna et al. 2017; Hudson and Manning 2019; Johnson et al. 2017; Acharya, Kafle, and Kanan 2018; Lin et al. 2014; Deitke et al. 2025).

Recent VLM counting studies also caution that benchmark design and response format can dominate the apparent conclusion. Diagnostic counting suites expose failures that are not visible in broad VQA scores (Guo et al. 2025; Hou et al. 2025; Alzahrani et al. 2026). Mechanistic and bottleneck-oriented studies argue that exact numerosity requires more than fluent visual description (Hasani et al. 2025; Che et al. 2026; Pang et al. 2026). Detector-grounded approaches offer a plausible repair path, but only when proposals or detections are actually available as scored evidence (Chen et al. 2026; Liu et al. 2023b; Kirillov et al. 2023).

The completed strict evidence does not support the hoped-for positive method claim. It instead supports a diagnostic result: exact counting remains weak across the current family mix, and CAPTURE real is the binding empirical weakness. The strict run records a scored-row count of 3360 and 7140 image requests. Overall exact match is 0.158631 with mean absolute error 16.582143. CAPTURE real contributes 2400 rows and falls to 0.040833 exact match with 22.422083 mean absolute error. PixMo-Count and TallyQA are stronger, with 0.471354 and 0.440972 exact match respectively, but they do not rescue a broad cross-family improvement story.

This negative result is still useful because it separates three claims that are often conflated. A completed run can be valid evidence without supporting the intended contribution. A decomposition condition can be a meaningful diagnostic probe without being a winning method. A family-level failure can be reported honestly without attributing it to an unmeasured visual property. The paper is organized around those separations. We first define the strict evaluation contract, then report the completed matrix, then use the matrix to reject unsupported improvement language.

The contribution is therefore deliberately narrower than a method-performance paper. We document the strict train-free evaluation, report the completed measured matrix, show

that numeric wrong answers dominate despite successful parsing, preserve fail-closed unavailable cells, and reject broad crop/grid improvement language. This framing makes the diagnostic evidence legible without converting weak results into a positive claim.

Related Work

Counting has long connected perception, attention, and exact enumeration. Classical numerosity studies explain why small counts can be perceived quickly while larger counts often require serial attention or approximation (Kaufman et al. 1949; Trick and Pylyshyn 1994; Jevons 1871; Piazza et al. 2004). Crowd and open-world counting systems instead emphasize density, localization, category grounding, and benchmark-specific annotation rules (Zhang et al. 2016; Ranjan et al. 2021; Liang et al. 2023; Amini-Naieni, Han, and Zisserman 2024; Chen et al. 2026).

Vision-language counting benchmarks vary in what makes a row difficult. VQA, GQA, CLEVR, COCO, Visual Genome, TallyQA, PixMo-Count, and CAPTURE-like settings all expose image-grounded numeric answering, but they differ in question wording, object categories, available metadata, local-count structure, and image access (Agrawal et al. 2016; Hudson and Manning 2019; Johnson et al. 2017; Lin et al. 2014; Krishna et al. 2017; Acharya, Kafle, and Kanan 2018; Deitke et al. 2025; Pothiraj et al. 2025). This matters for evidence discipline. Metadata without images is not a scored visual benchmark, and a benchmark with image-level counts does not automatically provide local-cell failure explanations.

Decomposition methods are the natural comparison class for this study. Chain-of-thought and zero-shot reasoning scaffolds change the response process without changing the visual evidence (Wei et al. 2022; Kojima et al. 2022). Local-view and divide-and-conquer strategies change the visual subproblem by asking a model to reason over crops or regions (Qharabagh, Ghofrani, and Fountoulakis 2024; Chen et al. 2026). Detector and segmentation tools such as Faster R-CNN, YOLO, DETR, GroundingDINO, and SAM show why object-aware comparisons require verified boxes, masks, or proposal systems rather than oracle annotations (Ren et al. 2017; Redmon et al. 2015; Carion et al. 2020; Liu et al. 2023b; Kirillov et al. 2023). The current strict package keeps that distinction visible by reporting detector/object-aware conditions as unavailable, not as scored baselines.

This work differs from a benchmark survey because it treats evidence availability as part of the result. Counting datasets expose different combinations of image access, object category wording, instance-count targets, and local annotations. A source with image-level answers can support exact-match evaluation, but it does not necessarily support claims about which image regions caused a miss. A source with metadata can help plan an evaluation, but metadata alone does not create a scored visual row. The strict package therefore reports each family according to what is actually executable under a shared train-free harness.

The same distinction applies to model comparisons. A second VLM backend, a detector-assisted decomposition, or an FSC-147 condition would each make the study stronger,

but only if the corresponding image-understanding backend, detector proposal source, or image-backed task rows are verified. Treating those cells as zero would confuse absence of evidence with a measured failure. Treating them as successful baselines would be worse. The current paper instead preserves them as fail-closed cells and limits the main evidence to conditions that produced scored rows.

Method and Experimental Setup

The strict evaluation is train-free: no supervised fine-tuning, reinforcement learning, preference optimization, detector tuning, learned scorer changes, or model-weight updates are part of the current evidence. Each ready task supplies an image, a natural-language counting prompt, and an official integer target. The evaluated system is one fixed hosted multimodal snapshot, used as one backend through the same image-counting harness across all executed conditions. The response contract asks for a single integer answer, generation is capped at 4096 output tokens, and no temperature or top-p override is used in the reported run. Raw responses are converted to integers by deterministic extraction, then scored with exact match and absolute error. CAPTURE-compatible sMAPE is retained in the aggregate tables where applicable.

Three benchmark families are ready and scored. CAPTURE real validation contributes 100 tasks, repeated across eight methods and three repeats for a scored-row count of 2400. TallyQA contributes 24 ready tasks and a scored-row count of 576. PixMo-Count contributes 16 ready tasks and a scored-row count of 384. Table 1 summarizes the completed family-level result.

The unit of analysis is a scored response row. Each row records the family, task identifier, method condition, repeat index, official count, parsed integer, exact-match indicator, absolute error, and parse/empty-response flags. Repeats are retained as separate rows rather than averaged away before scoring, so a method that is unstable across repeated calls is penalized directly in the row-level exact-match rate. The family aggregates in Table 1 are therefore simple row means over the completed strict matrix, not hand-selected examples.

The three ready families stress different parts of the counting problem. CAPTURE real has much larger targets in the accepted rows, with official counts ranging from 9 to 672. PixMo-Count is smaller in this package, with official counts from 3 to 10. TallyQA spans 1 to 15 in the prepared subset. This makes the overall score intentionally difficult to interpret as a single capability number. The aggregate exact match is useful as a summary of the strict matrix, but the family rows are the primary diagnostic evidence.

CAPTURE real is therefore more than another row in the average. It is the portion of the matrix where the official counts most directly stress high-cardinality exact enumeration. The evaluated prompts usually return parseable integers, but the returned integers are far from the target often enough that the family-level mean absolute error exceeds twenty. This is why the paper treats CAPTURE real as the binding empirical weakness. A method that improves only

Family	Rows	EM	MAE
CAPTURE real	2400	0.040833	22.422083
PixMo-Count	384	0.471354	2.065104
TallyQA	576	0.440972	1.927083
Overall	3360	0.158631	16.582143

Table 1: Completed strict evidence is diagnostic, with a scored-row count of 3360 and weak cross-family exact counting. CAPTURE real is the binding weakness at 0.040833 exact match and 22.422083 mean absolute error.

the smaller-count families would not solve the strict evaluation as declared.

PixMo-Count and TallyQA play a different role. They show that the evaluated system can sometimes answer counting questions exactly under this harness, which prevents the negative result from being dismissed as a broken parser or broken API path. Their stronger full-image conditions also make the crop/local failures more informative: the local conditions are not merely tied with all other methods in small-count settings. They actively underperform conditions that keep the whole image visible and ask for a direct or lightly scaffolded answer.

The train-free constraint is intentionally strict because otherwise the comparison would change character. A learned detector, tuned prompt router, or fine-tuned counting head could be a valid future method, but it would no longer be the same zero-update diagnostic. In this paper, every condition is an inference-time intervention. The cost of that choice is lower ceiling performance; the benefit is that the evidence isolates how far the evaluated model and simple decomposition prompts get without training or fitted components.

The eight scored train-free conditions are direct answer, chain-of-thought, grid-only prompting, LVLM-style local decomposition, crop-grid tally, crop without point prompts, full-image point-style prompting, and final-numeral extraction. These conditions are useful as diagnostic probes, but the completed matrix does not show a broad positive crop/grid improvement. CAPTURE real remains near floor for every condition, and the crop/local conditions are mixed or poor on the other two families. The safest claim is therefore not that one train-free method wins, but that the strict matrix exposes weak exact counting and unstable behavior across source families.

The method conditions are deliberately lightweight. Direct answering asks for a single integer from the full image. Chain-of-thought changes only the response scaffold. Grid-only prompting asks the model to reason over a spatial partition while still using the full image. LVLM-style local decomposition and crop-grid tally expose local views or local counting structure, then aggregate the response into a final number. Crop without points removes point-style cues from the crop condition. Full-image point-style prompting asks for point-like counting behavior without cropping. Final-numeral extraction tests whether using the last produced number changes scoring. None of these conditions updates model weights or introduces a learned detector.

This design also explains why the paper does not run a paired-significance table for a positive improvement claim. The proposed crop/local family does not consistently beat the relevant full-image conditions, so there is no positive effect to elevate with an inferential test. The analysis therefore reports descriptive diagnostic comparisons and leaves paired tests for a future method claim that is actually supported by the measured matrix.

The scoring semantics are intentionally conservative. Exact match is the primary metric because the application-facing question asks for one integer and the official targets are integers. Mean absolute error is reported because it distinguishes near misses from severe misses, especially when larger counts dominate a family. sMAPE is retained for CAPTURE-compatible summaries, but it is not used to rescue a condition that fails exact match. This metric ordering prevents a method from being described as successful only because it reduces large errors while still returning the wrong count most of the time.

The parser policy is also conservative. The harness extracts a final integer from the raw response and records whether parsing failed or the response was empty. In the completed strict run, those failure flags are zero across all 3360 scored rows. That fact is important because it moves the diagnostic away from output-format cleanup. The run did not need a more permissive regular expression to recover hidden correct answers. It needed better visual counting.

Results

Figure 1 shows the main result. Exact match is source-dependent and generally weak under the strict setting. CAPTURE real is the clearest collapse: the best family-level condition in the matrix remains low, and the family aggregate is 0.040833 exact match. TallyQA and PixMo-Count have higher exact match for some full-image or point-style conditions, but local decomposition does not produce a stable cross-family gain.

Mean absolute error tells the same story with a different scale. The lower panel of Figure 1 shows large CAPTURE real errors and lower errors on TallyQA and PixMo-Count. The family imbalance matters: CAPTURE real contributes most scored rows and dominates the aggregate error, so a positive overall claim would need a method that substantially improves this family. The current completed rows do not provide that evidence.

Collapsing across families makes the same point from the method side. Direct, chain-of-thought, grid-only, full-image point-style prompting, and final-numeral extraction all land in a narrow overall exact-match band between 0.209524 and 0.247619. The crop/local conditions are worse overall: LVLM-style local decomposition reaches 0.045238 exact match, crop-grid tally reaches 0.040476, and crop without points reaches 0.033333. These aggregate rows should not be read as a universal method ranking because the family mix is uneven, but they are enough to reject a broad crop/grid win under the declared evaluation.

The failure taxonomy rules out a common but incorrect explanation. The run did not fail because responses were empty or unparsable. Across 3360 rows, parse failures are

Source	Tasks/split	Direct EM	Best full-image EM	Crop-grid EM	Family EM
CAPTURe real	100 real-validation	0.053333	0.063333	0.046667	0.040833
PixMo-Count	16 test	0.687500	0.854167	0.000000	0.471354
TallyQA	24 simple split	0.722222	0.722222	0.041667	0.440972

Table 2: Cross-benchmark diagnostic matrix for the completed strict evaluation. The same fixed hosted VLM snapshot, 4096-token output cap, three-repeat policy, deterministic integer extraction, and exact-match metric are used across sources; crop-grid decomposition does not support a broad improvement claim.

zero and empty responses are zero. The dominant failure type is parsed numeric wrong answer: 2827 rows return a number that does not match the official count. Figure 2 makes this failure mode explicit.

The wrong-answer counts are concentrated where the count range is larger, but they are not limited to one family. CAPTURe real has 2302 wrong numeric rows out of 2400. PixMo-Count has 203 wrong numeric rows out of 384, and TallyQA has 322 out of 576. Because every response row produced a parseable number, the dominant defect is not a formatting failure that could be fixed by a more permissive parser. The observed defect is that the model often returns the wrong integer under the strict prompt and image conditions.

The result also shows that local decomposition is not automatically safer than full-image prompting. The intuition behind crops and grids is reasonable: a smaller view may reduce clutter, and summing local counts might be easier than counting a whole image. The strict matrix tests that intuition and finds that it does not hold in the current implementation. Local methods can lose global context, double-count boundary objects, miss instances that are split across cells, or amplify small local errors through aggregation. The present evidence does not isolate which of those explanations dominates, but it does show that the simple crop/local variants are not enough.

Analysis

The current evidence supports three claims. First, strict train-free visual counting is weak in this benchmark mix, with overall exact match 0.158631. Second, CAPTURe real is the binding empirical weakness, with 0.040833 exact match and 22.422083 mean absolute error. Third, the failure mode is numeric miscounting after successful parsing, not parser collapse. These claims are derived from the scored rows and aggregate tables.

The evidence rejects the intended broad positive method claim. A crop/grid method can only be reported as an improvement if it improves the relevant comparator under the declared family mix. The current matrix does not meet that standard. On CAPTURe real, crop-grid tally has 0.046667 exact match, below direct prompting at 0.053333 and below grid-only prompting at 0.063333. On PixMo-Count, crop-grid tally is 0.000000 while several full-image conditions are much stronger. On TallyQA, crop-grid tally is 0.041667, far below direct prompting at 0.722222. These values make broad crop/grid improvement language unsupported.

The strongest local view of the data is family-specific, not method-general. On CAPTURe real, all conditions are

near floor: grid-only prompting is the highest exact-match condition at 0.063333, while full-image points is 0.006667. On PixMo-Count, the best condition is full-image points at 0.854167, followed by chain-of-thought at 0.770833, while the crop/local conditions score 0.000000. On TallyQA, direct answer is strongest at 0.722222, while crop-grid tally and crop without points both score 0.041667. A method paper would need to explain a robust improvement pattern; the present matrix shows the opposite.

The mean-absolute-error rows make the failure less binary. CAPTURe real has high error under every condition, but crop-grid tally has lower mean absolute error than direct answer on that family despite lower exact match. That pattern suggests that some local decomposition prompts may move predictions closer without making them exactly correct. The current paper does not promote this as a win because exact match is the declared primary metric and because the same crop/local conditions are poor on PixMo-Count and TallyQA. It is instead a useful diagnostic observation for future method design.

True-count bins further clarify the diagnostic character of the result. Figure 3 shows that larger counts are harder, especially in CAPTURe real. This is consistent with exact-count difficulty, but it should not be overread as a mechanism proof: the current scored rows do not include verified occlusion or density metadata. The correct paper-facing interpretation is a numerical error pattern, not an attributed visual-cause claim.

The row-derived count bins quantify that pattern. Rows with official counts from 0 to 5 have 0.566288 exact match and 1.628788 mean absolute error. Counts from 6 to 10 fall to 0.317982 exact match and 2.592105 mean absolute error. Counts from 11 to 25 fall further to 0.070736 exact match and 5.546512 mean absolute error. Counts from 26 to 50 reach 0.014423 exact match and 11.498397 mean absolute error, and counts above 50 reach 0.009722 exact match with 56.631944 mean absolute error. This monotone degradation is the clearest mechanism-adjacent result that the current rows can support.

The same bins also show why the paper avoids occlusion and density explanations. CAPTURe real contains large-count rows and is designed around occluded-object counting, so it is tempting to attribute failure to occlusion. The accepted scored-row schema, however, does not contain verified row-level occlusion or density fields. The paper can say that large counts are difficult and that CAPTURe real collapses in this package. It cannot say which visual scene property caused each miss without adding verified metadata or a separate analysis.

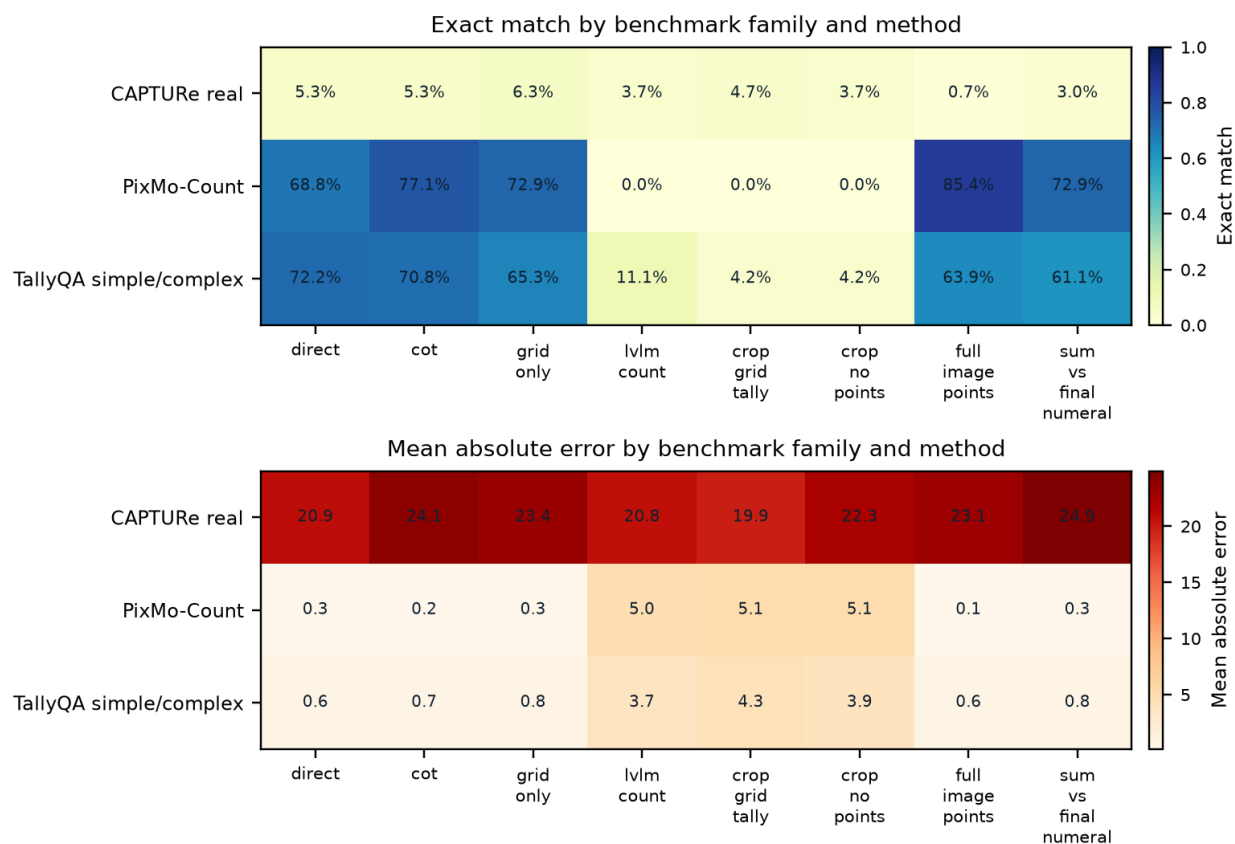


Figure 1: Exact match and mean absolute error by benchmark family and train-free condition from the completed strict scored rows. The panels support a diagnostic interpretation: CAPTURE real remains weak across methods, larger errors concentrate in that family, and no broad crop/grid improvement claim is supported.

There is also a distinction between rejecting the current method claim and rejecting all decomposition work. The current evidence rejects the implemented train-free crop/grid variants under the declared strict package. It does not prove that every local or detector-assisted method must fail. In fact, the fail-closed object-aware condition marks exactly the stronger comparison that would be needed to test a more grounded decomposition story. A verified detector pipeline could change the empirical picture; until it exists as scored evidence, it remains an open comparison rather than a paper result.

The negative result therefore has a constructive interpretation. It identifies where future work should spend effort: high-count CAPTURE-like rows, detector-quality object proposals, cross-backend replication, and row-level visual metadata for mechanism claims. These directions are not added as hidden success claims. They follow from the measured failure surface and the fail-closed cells.

Fail-Closed Conditions

Three planned comparison cells remain fail-closed rather than scored. FSC-147 is blocked because the local preparation has metadata but not verified image-backed rows. A secondary VLM condition is blocked because a second

image-understanding backend was not verified for this strict run. The object-aware detector/decomposition condition is blocked because detector weights or a detector route were not verified; schema fixtures do not count as a usable detector comparator.

Fail-closed reporting is part of the scientific result because it prevents scope inflation. A paper could look broader by silently dropping unavailable cells or by treating metadata and fixtures as if they were scored visual conditions. This study does neither. It reports the completed scored matrix for the available families and marks the unavailable cells as evidence gaps.

Each fail-closed cell blocks a different generalization. The missing secondary VLM condition blocks backend-general language; the results are evidence about one evaluated hosted multimodal system, not about all VLMs. The missing FSC-147 image-backed rows block a claim that the strict matrix covers few-shot exemplar counting with that dataset. The missing object-aware detector condition blocks claims that detector-guided decomposition has been fairly tested. These limitations are not cosmetic. They define the boundary between the current diagnostic contribution and the stronger experiment that would be needed for an AAAI positive-method paper.

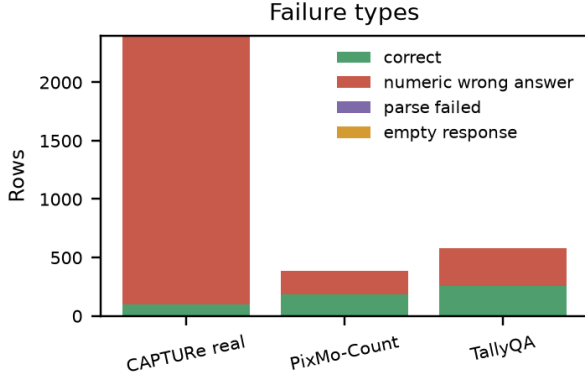


Figure 2: Failure types from completed scored rows. Parser and empty-response failures are absent; wrong numeric answers dominate the strict run.

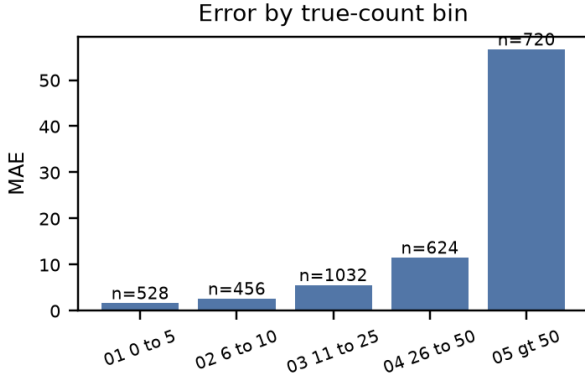


Figure 3: Mean absolute error by true-count bin from completed scored rows. Error grows with harder count ranges, but occlusion and density metadata are unavailable in the scored-row schema.

The fail-closed cells also shape the correct baseline language. The paper can compare among the eight executed prompt/decomposition conditions because those rows exist under the same primary backend. It cannot compare against an object-aware detector pipeline as if fixture schemas or planned proposals were scored outputs. It also cannot claim robustness across model providers. The honest statement is that the available matrix is complete for the ready families and executed conditions, while the broader comparison matrix remains open.

The same policy prevents a misleading treatment of FSC-147. FSC-147 is highly relevant to visual counting, and including it would strengthen the paper if image-backed rows were available under the same scoring contract. But a family cannot be included because its name is relevant. It must have accessible images, verified targets, and task rows that the harness actually scores. The present manuscript there-

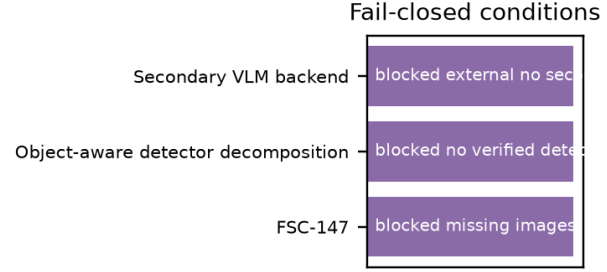


Figure 4: Unavailable strict conditions are preserved as fail-closed cells. FSC-147, secondary VLM, and object-aware detector/decomposition are not assigned fabricated scores.

fore treats FSC-147 as a missing condition rather than an omitted negative result.

Claim Boundary

Table 3 records the claim boundary used in the paper. Supported claims are diagnostic and source-scoped. Unsupported claims are omitted or softened. The most important omitted claim is the broad statement that crop/grid decomposition improves exact visual counting. The current evidence instead says that train-free exact counting is weak under this strict setup, that CAPTURE real is the hardest ready family, and that wrong numeric answers dominate.

This boundary also protects interpretation of negative results. The current matrix is sufficient to reject a broad positive improvement claim, but it is not sufficient to explain every visual mechanism behind the failures. For example, the rows show high absolute errors in large-count bins, yet they do not contain enough verified occlusion or density metadata to attribute the failures to a specific scene property. The paper therefore keeps mechanism language conservative.

A second boundary concerns contribution language. The current paper is coherent as a negative diagnostic study because it has real scored rows, current tables, and figures tied to the completed evaluation. It is not a broad positive method-improvement paper because the central intended contribution was not observed. This distinction matters for peer review: the paper should be judged on whether the diagnostic result is worth publishing, not on a hidden claim that the method wins.

A third boundary concerns evaluation scope. The paper uses the completed strict evaluation as the source of scored claims. Other attempted or planned cells are not mixed into the current evidence matrix unless they are executed under the same scoring protocol.

Discussion

The main lesson is methodological. Train-free visual counting evaluations can look deceptively complete when a table contains many prompt variants, but the scientific claim depends on the relationship between rows. A broad improvement claim needs a consistent advantage over relevant base-

Claim	Current disposition	Evidence	Manuscript action
Broad crop/grid improvement	rejected	completed family-method matrix	omitted as positive claim
Weak strict visual counting	supported	a scored-row count of 3360; EM 0.158631	used as main diagnostic claim
CAPTURe real collapse	supported	2400 CAPTURe rows; EM 0.040833	highlighted as binding weakness
Parser collapse	rejected	zero parse and empty failures	replaced with numeric-error claim
Secondary VLM comparison	unavailable	fail-closed backend cell	not scored
FSC-147 coverage	unavailable	missing image-backed rows	not scored
Object-aware detector comparator	unavailable	no verified detector resource	not scored

Table 3: Claim boundary for the completed strict evidence. The paper reports diagnostic failures and unavailable cells, not a positive method-improvement result.

lines, preferably across families and with a strong comparator. The completed strict matrix does not have that shape. It has a reliable negative shape: all conditions struggle on CAPTURe real, and the crop/local conditions do not compensate by winning elsewhere.

This changes how the paper should be read. The useful contribution is not a new algorithm but a disciplined diagnostic record. It shows how a planned positive claim can fail under stricter evidence without being hidden by prose. It also shows how to preserve missing comparisons without converting them into zeros or successes. That may be less exciting than a winning method, but it is a meaningful result for a field where counting claims can otherwise be inflated by small subsets, unverified local annotations, or incomplete comparator matrices.

The strict package also raises a benchmark-design issue. A single exact-match aggregate can combine very different count regimes. In this run, the smallest-count rows are much easier than rows above 50, and CAPTURe real has the widest target range. Future evaluations should report count bins, family-specific metrics, and method-specific failure modes by default. Without those slices, a model can appear moderately competent on aggregate while failing exactly where high-cardinality counting is hardest.

Another lesson concerns response scaffolding. Chain-of-thought and grid prompting are sometimes treated as general-purpose improvements, but this matrix does not support that broad view. Their behavior is family-dependent, and none of the scaffolds removes the CAPTURe real failure. Full-image point-style prompting is strong on PixMo-Count but weak on CAPTURe real, while direct answering remains strongest on TallyQA. These mixed outcomes suggest that prompt form interacts with dataset structure rather than providing a general counting solution.

Finally, the negative result clarifies what would count as progress. A future paper could improve this one by adding a verified second backend, running a detector-assisted object-aware condition, preparing image-backed FSC-147 rows, and attaching row-level occlusion or density metadata. A future method could also be judged by whether it improves CAPTURe real exact match without sacrificing the smaller-count families. Those are concrete requirements derived from the current failure surface, not generic calls for more experiments.

Limitations

The primary limitation is that the evidence is negative and diagnostic. It does not provide a new winning method for exact visual counting. That limitation is substantive rather than stylistic: the completed matrix contradicts the broad crop/grid improvement story that originally motivated the evaluation. A future version would need either a stronger method, a stronger comparator, or a narrower target claim.

The family mix is also uneven. CAPTURe real supplies most rows, while TallyQA and PixMo-Count are smaller scored sources. This makes CAPTURe real the binding empirical weakness and makes the overall exact-match average sensitive to that family. The paper therefore reports family-level values alongside the aggregate rather than asking the reader to infer all behavior from one number.

The current rows lack verified occlusion and density metadata, so detailed visual-mechanism claims remain out of scope. The true-count bins show that larger counts are harder, and the CAPTURe real subset shows severe error, but those observations do not identify the visual cause of each mistake. Finally, secondary VLM, FSC-147, and object-aware detector/decomposition cells are fail-closed rather than scored, so backend generality and detector-assisted comparisons remain open.

There are also limits in statistical interpretation. The paper reports repeated responses and descriptive row-level summaries, but it does not claim a significant positive method difference because the underlying direction is not favorable to the intended method. This is a conservative choice. It avoids presenting a formal test as if it could turn a mixed or negative matrix into a positive contribution.

The evaluated system is one fixed hosted multimodal snapshot under one primary backend. The results should not be generalized to every current or future VLM without replication. The failure may be milder or worse for other models, and a detector-assisted method might behave differently. The evidence is still useful because it is complete for the declared ready families and executed conditions, but its scope is the strict evaluation reported here.

Ethical Considerations

This study does not train a model or deploy a counting system. The main ethical risk is misrepresenting reliability: ex-

act counts may be used in settings where numeric errors matter, and a paper that overstates visual counting performance could encourage unsafe reliance on brittle model outputs. The fail-closed policy reduces that risk by reporting unavailable conditions and negative measured outcomes directly.

The negative result also matters for responsible communication. A reader should not leave with the impression that decomposition made counting reliable. The measured errors are large enough that exact-count outputs from the evaluated setting should not be used as authoritative counts in safety-critical, labor, inventory, ecological, medical, or accessibility contexts without task-specific validation. The paper’s conservative claim boundary is therefore part of the ethical posture, not only a review tactic.

Reproducibility

The evaluation can be replayed from the scored response rows, parsed integers, official targets, and aggregate result tables. Each response row records benchmark family, task identifier, method condition, repeat index, predicted integer, target integer, exact-match value, absolute error, and parse/empty-response flags. The reported family and method summaries are row means over those completed responses.

The replay contract has three invariants. First, the scored rows from the completed strict evaluation define the numeric claims in this paper. Second, every reported number in the main results is derived from row-level scores or aggregate tables. Third, unavailable cells remain unavailable until the required images, backend, or detector resource has been verified and scored under the same harness. These invariants make the negative result checkable even though the paper is not a positive method paper.

The evaluation also separates raw response parsing from scoring. A response is converted to an integer by deterministic extraction and then compared with the official target. Because the current run has no parse failures and no empty responses, changing the parser would not explain the dominant error pattern. A future replication should therefore preserve both the raw response and the parsed integer so that parser behavior and counting behavior remain distinguishable.

For future replications, the most important rule is that claims follow the scored rows. If new scored cells are added, the aggregate tables and claim boundary should be recomputed from those rows. If no new evidence is added, the current negative boundary should remain in place.

The negative-result boundary is also reproducible. A reader can check it by asking whether the executed crop/local rows beat the relevant full-image rows across the declared families. They do not: CAPTURE real stays near floor, PixMo-Count favors full-image point-style or chain-of-thought prompting, and TallyQA favors direct answering. This makes the claim decision independent of author preference. The paper does not need a subjective judgment that the method is disappointing; the table values are enough to rule out the broad improvement sentence.

The same replay logic applies to fail-closed cells. If a later study verifies a second backend, prepares FSC-147

image-backed rows, or runs an object-aware detector condition, those cells should enter as new scored rows under a clearly stated evaluation setting. They should not be merged silently into the completed strict run reported here. Keeping that boundary explicit leaves a clear path for stronger future evidence.

Conclusion

The completed strict evidence supports a train-free visual numerosity diagnostic study, not a positive method paper. Across 3360 scored rows and 7140 image requests, overall exact match is 0.158631. CAPTURE real collapses to 0.040833 exact match and 22.422083 mean absolute error. There are no parse or empty-output failures, while 2827 rows are wrong numeric answers. Secondary VLM, FSC-147, and object-aware detector/decomposition conditions remain fail-closed.

The practical conclusion is the claim boundary. The implemented crop/local variants do not support a broad exact-counting improvement claim, and unavailable cells cannot be promoted into scored baselines. What the paper can report is narrower but coherent: strict train-free exact counting remains weak for the evaluated hosted multimodal snapshot, CAPTURE real is the binding family, and wrong numeric answers dominate the failure taxonomy.

This boundary gives future work a concrete target. A stronger positive submission would need a verified second backend, a scored object-aware comparator, image-backed FSC-147 rows, or a method that improves the CAPTURE real failure without damaging smaller-count families. Until that evidence exists, the contribution is the diagnostic record itself: it separates what was measured from what remains to be tested.

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Supplementary Replay Notes

The supplementary materials contain row-level scored outputs, parsed integers, official targets, aggregate result tables, and failure-type summaries for the completed strict evaluation. They are sufficient to recompute the paper’s family-level metrics, method summaries, and parse/empty-response counts. The body reports the method-level diagnostic pattern in prose and in Figure 1, so the appendix does not duplicate those aggregates as a separate table.